Hints: Limits – 3. Asymptotics of sequences

Homework

The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.

Problem 1

Hint 1. The map $g(x) = x - \arcsin(\sin^3 x)$ is odd, so it suffices to take $x_0 \in (0, 1]$ (for $x_0 = 0$ the sequence is 0). For $x \in (0, 1]$: $0 < \sin^3 x < \sin x$, so $0 < \arcsin(\sin^3 x) < x$; hence x_n stays in $(0, 1]$, decreases, and tends to a fixed point, i.e. to 0.

Hint 2. Here $\arcsin(\sin^3 x) = x^3 + O(x^5)$, so $x_{n+1} = x_n - x_n^3 + O(x_n^5)$: Theorem 1 with $p = 3$, $a = 1$. Telescoping: $\dfrac{1}{x_{n+1}^2} - \dfrac{1}{x_n^2} \to 2$, and Stolz gives $nx_n^2 \to \frac{1}{2}$.

Hint 3. Answer: $\lim \sqrt{n}\, x_n = \frac{1}{\sqrt{2}}$ for $x_0 \in (0, 1]$, $-\frac{1}{\sqrt{2}}$ for $x_0 \in [-1, 0)$, and 0 for $x_0 = 0$.

Problem 2

Hint 1. Substitute $t = nx$: $b_n = \frac{1}{n}\int_0^n f(t)\,dt = \frac{1}{n}\sum_{k=0}^{n-1} a_k$.

Hint 2. So b_n is the Cesàro mean of (a_k), and converges to the same limit (Stolz–Cesàro).

Problem 3

Hint 1. The product equals $P_n = \dfrac{(n!)^2}{(2n)!}$, and the exponent is $e^{1999/n} - 1 = \dfrac{1999}{n}\,(1 + o(1))$. So the logarithm of the expression is $1999 \cdot \dfrac{\ln P_n}{n}\,(1 + o(1))$.

Hint 2. Find $\lim \frac{\ln P_n}{n}$ by Stolz: it equals $\lim(\ln P_{n+1} - \ln P_n) = \lim \ln \dfrac{(n+1)^2}{(2n+1)(2n+2)} = -\ln 4$. (Stirling gives more, $\ln P_n = -n \ln 4 + \frac{1}{2}\ln(\pi n) + o(1)$, but it is not needed.)

Hint 3. The logarithm of the expression tends to $-1999 \ln 4$. Answer: 4^{-1999}.

Problem 4

Hint 1. The numerator is a geometric progression: $\frac{n^{n+1}-n}{n-1} \sim n^n$. Divide everything by n^n.

Hint 2. The denominator becomes $\sum_{j=0}^{n-1}(1 - \frac{j}{n})^n$. Each term tends to e^{-j} and is at most e^{-j} (since $1 - t \leqslant e^{-t}$), so the sum tends to $\sum_{j \geqslant 0} e^{-j} = \frac{e}{e-1}$ (cut the sum at a large fixed J; the tail is at most $\sum_{j>J} e^{-j}$).

Hint 3. Answer: $\dfrac{e-1}{e}$.

Problem 5

Hint 1. a) $g_n(x) = f(x) - x^n$ is strictly decreasing, $g_n(0) = f(0) > f(1) \geqslant 0$ and $g_n(1) = f(1) - 1 < f(0) - 1 \leqslant 0$: one root $a_n \in (0, 1)$. Since $g_{n+1}(a_n) = a_n^n - a_n^{n+1} > 0$, the sequence a_n increases. If $a_n \to L < 1$, then $a_n^n \to 0$ and $f(L) = 0$, but $f(1) < f(L)$ and $f(1) \geqslant 0$.

Hint 2. b) Put $t_n = 1 - a_n \to 0$ and $c = -f'(1) > 0$. Then $f(a_n) = ct_n(1 + o(1))$ and $a_n^n = e^{-nt_n(1+o(1))}$, so taking logarithms,

$$nt_n(1 + o(1)) = \ln \frac{1}{t_n} + O(1).$$

Deduce $t_n \sim \frac{\ln n}{n}$ (first show $\frac{1}{n} \lesssim t_n \lesssim n^{-1/2}$, so that $\ln \frac{1}{t_n} \sim \ln n$).

Hint 3. $F(a_n) = \int_{a_n}^{1} f \sim \frac{c}{2} t_n^2 \sim \frac{c}{2}\left(\frac{\ln n}{n}\right)^2$. So $\sum F(a_n)^\alpha$ behaves like $\sum \left(\frac{\ln n}{n}\right)^{2\alpha}$: it converges exactly for $\alpha > \frac{1}{2}$ (for $\alpha \leqslant 0$ the terms do not even tend to 0).

Problem 6

Hint 1. Rewrite: $x_{n+2} - 2x_{n+1} = (x_{n+1} - 2x_n) + \frac{2^n}{x_n}$. So $d_n = x_{n+1} - 2x_n$ satisfies $d_1 = 0$, $d_{n+1} = d_n + \frac{2^n}{x_n}$.

Hint 2. Put $y_n = x_n/2^n$. Dividing by 2^{n+1}: $y_{n+1} - y_n = \frac{d_n}{2^{n+1}}$, and $\frac{2^n}{x_n} = \frac{1}{y_n}$, so $d_n = \sum_{k=1}^{n-1} \frac{1}{y_k}$. Hence $d_n \geqslant 0$, y_n is non-decreasing and $y_n \geqslant y_1 = 1$.

Hint 3. Sum the increments and swap the order of summation, using $\sum_{n=k+1}^{N-1} 2^{-n-1} = 2^{-k-1} - 2^{-N}$:

$$y_N = 1 + \sum_{n=1}^{N-1} \frac{1}{2^{n+1}} \sum_{k=1}^{n-1} \frac{1}{y_k} = 1 + \sum_{k=1}^{N-2} \frac{2^{-k-1} - 2^{-N}}{y_k}.$$

Since $y_k \geqslant 1$, $y_N < 1 + \sum_{k \geqslant 1} 2^{-k-1} = \frac{3}{2}$: the sequence is non-decreasing and bounded, so $L = \lim y_n$ exists, and letting $N \to \infty$,

$$L = 1 + \sum_{k \geqslant 1} \frac{1}{2^{k+1} y_k}.$$

Hint 4. Now use $y_k \leqslant L$. Already $y_k \in [1, L]$ gives $1 + \frac{1}{2L} \leqslant L \leqslant \frac{3}{2}$, i.e. $\frac{1+\sqrt{3}}{2} \leqslant L \leqslant \frac{3}{2}$. Sharper: $y_1 = y_2 = 1$, $y_3 = \frac{9}{8}$, and $y_k \in [\frac{9}{8}, L]$ for $k \geqslant 3$, so

$$\frac{11}{8} + \frac{1}{8L} \leqslant L \leqslant \frac{11}{8} + \frac{1}{9}, \qquad \text{i.e.} \qquad 1.460 < \frac{11 + \sqrt{153}}{16} \leqslant L \leqslant \frac{107}{72} < 1.487.$$

Numerically $L \approx 1.4782$.

Problem 7

Hint 1. Use $\frac{2x}{\pi} \leqslant \sin x \leqslant x$ on $[0, \frac{\pi}{2}]$. a) Near 0 the integrand is $\frac{\{nx\}}{\sin x} \leqslant \frac{nx}{\sin x} \leqslant \frac{\pi n}{2}$, so it is bounded and the integral converges. b) The same inequality gives $J(n) \leqslant I(n) \leqslant \frac{\pi}{2} J(n)$, where, after $t = nx$,

$$J(n) = \int_0^{\pi/2} \frac{\{nx\}}{x}\, dx = \int_0^{n\pi/2} \frac{\{t\}}{t}\, dt.$$

So it suffices to show that $J(n)$ is of order $\ln n$.

Hint 2. Upper bound: $\{t\} \leqslant \min(t, 1)$, so $J(n) \leqslant \int_0^1 dt + \int_1^{n\pi/2} \frac{dt}{t} = 1 + \ln \frac{\pi n}{2}$. Lower bound: on each $[k, k+1]$, $\int_k^{k+1} \frac{\{t\}}{t}\, dt = \int_0^1 \frac{s}{k+s}\, ds \geqslant \frac{1}{2(k+1)}$, and $[1, n] \subset [0, \frac{n\pi}{2}]$, so

$$J(n) \geqslant \sum_{k=1}^{n-1} \frac{1}{2(k+1)} = \frac{H_n - 1}{2}.$$

Hint 3. For $n \geqslant 2$: $H_n - 1 = \sum_{j=2}^{n} \frac{1}{j} \geqslant \int_2^{n+1} \frac{dt}{t} = \ln \frac{n+1}{2} \geqslant \frac{1}{2} \ln n$ (because $\frac{n+1}{2} \geqslant \sqrt{n}$), and $1 + \ln \frac{\pi}{2} < 1.5 < 3 \ln 2 \leqslant 3 \ln n$. Hence $\frac{1}{4} \ln n \leqslant J(n) \leqslant 4 \ln n$, and

$$\frac{1}{4} \ln n \leqslant I(n) \leqslant 2\pi \ln n \qquad (n \geqslant 2).$$

(More precisely, $I(n) = \frac{1}{2} \ln n + O(1)$.)

Problem 8

Hint 1. Invert: $\dfrac{1}{a_{n+1}} = \dfrac{1}{a_n} + 1 + a_n$.

Hint 2. Hence $\frac{1}{a_n} \geqslant \frac{1}{a_0} + n$, so $a_n \to 0$; then $\frac{1}{a_{n+1}} - \frac{1}{a_n} \to 1$ and Stolz gives $\lim n a_n = 1$.

Problem 9

Hint 1. Write $1 + x + \cdots + x^n = \frac{1-x^{n+1}}{1-x}$, so $\ln(1 + \cdots + x^n) = \ln(1 - x^{n+1}) - \ln(1 - x)$. Since $\int_0^1 \ln^2(1-x)\,dx = 2$, you get, with $m = n + 1$,

$$2 - x_n = \int_0^1 \Big(-\ln(1 - x^m)\Big)\Big(-\ln(1 - x)\Big)dx \geqslant 0,$$

and the integrand is $O\big(\ln^2(1-x)\big)$ near 1, so everything is finite.

Hint 2. Expand $-\ln(1 - x^m) = \sum_{k\geqslant 1} \frac{x^{mk}}{k}$ and use $\int_0^1 x^N\big(-\ln(1-x)\big)dx = \frac{H_{N+1}}{N+1}$ (expand $-\ln(1-x)$ as well). This gives

$$2 - x_n = \sum_{k\geqslant 1} \frac{H_{mk+1}}{k\,(mk+1)}.$$

Hint 3. With $H_N = \ln N + O(1)$: $2 - x_n = \frac{1}{m}\sum_{k\geqslant 1} \frac{\ln m + \ln k + O(1)}{k^2} = \zeta(2)\frac{\ln m}{m} + O\big(\frac{1}{m}\big)$. In particular $x_n \to 2$, and $\lim \frac{n}{\ln n}(2 - x_n) = \zeta(2) = \frac{\pi^2}{6}$.

Problem 10

Hint 1. Write $\sqrt[n]{1 + x^n} - 1 = e^u - 1$ with $u = \frac{\ln(1+x^n)}{n} \in [0, \frac{\ln 2}{n}]$. Since $u \leqslant e^u - 1 \leqslant u + u^2$ for small u and $u \leqslant \frac{x^n}{n}$, the quadratic part contributes at most $n^2 \int_0^1 \frac{x^{2n}}{n^2}dx = \frac{1}{2n+1} \to 0$.

Hint 2. So it suffices to find $\lim n \int_0^1 \ln(1 + x^n)\,dx$. Substitute $x = t^{1/n}$: $n\int_0^1 \ln(1 + x^n)\,dx = \int_0^1 \frac{\ln(1+t)}{t}\,t^{1/n}\,dt$.

Hint 3. As $n \to \infty$ this tends to $\int_0^1 \frac{\ln(1+t)}{t}\,dt$ (the integrand increases to the limit). Expanding $\ln(1 + t)$, the integral equals $\sum_{k\geqslant 1} \frac{(-1)^{k+1}}{k^2} = \frac{\pi^2}{12}$.